

COSIMA Cookbook: a computational learning module for analysing ocean and sea-ice model output

Navid C. Constantinou^{1,11*}, Julia Neme^{2*}, Wilton Aguiar², Matthis Auger¹³, Ashley J. Barnes^{4,8}, Romain Beucher³, Dhruv Bhagtani², Christopher Yit Sen Bull⁴, Hannah Dawson¹³, Noah Day¹¹, Fabio Boeira Dias¹⁰, Edward Doddridge¹³, Anupiya Ellepola², Matthew England¹⁰, Denisse Fierro-Arcos¹³, Angus Gibson², Aidan Heerdegen³, Andy McC. Hogg^{3,4}, Ryan M. Holmes⁵, Wilma Huneke², Jemma Jeffree^{2,4}, Andrew E. Kiss², Minghang Li³, Josué Martínez-Moreno⁹, Jan Jaap Meijer¹³, Thomas Moore⁷, Ruth Moorman⁶, Adele Morrison², James Munroe¹, Aditya Narayanan¹², Micael Oliveira³, Ellie Q. Y. Ong^{4,8}, Max Proft³, Madelaine G. Rosevear^{4,13}, Christina Schmidt⁹, Taimoor Sohail¹¹, Paul Spence^{4,13}, Dougal T. Squire³, Anton Steketee³, Charles Turner³, Felipe Vilela da Silva¹³, Marc White³, Luwei Yang², Claire Yung², and Jan Zika¹⁰

¹ 2i2c ² Australian National University, Australia ³ Australian National University, Australia's Climate Simulator (ACCESS-NRI), Australia ⁴ Australian Research Council Centre of Excellence for the Weather of the 21st Century, Australia ⁵ Bureau of Meteorology, Australia ⁶ California Institute of Technology, USA ⁷ Commonwealth Scientific and Industrial Research Organisation, Australia ⁸ Monash University, Australia ⁹ British Antarctic Survey, UK ¹⁰ University of New South Wales Sydney, Australia ¹¹ University of Melbourne, Australia ¹² University of Southampton, UK ¹³ University of Tasmania, Australia * These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright²⁹ and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

The COSIMA Cookbook is an open computational learning module for analysing ocean and sea-ice model output in Jupyter notebooks. It has been developed by the Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA; <https://cosima.org.au>) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean-sea ice model configuration ([Kiss et al., 2020](#)). The repository combines introductory tutorials with worked analysis examples, all exposed through a browsable Sphinx documentation site ([Figure 1](#)) and backed by a citable archived release ([COSIMA contributors, 2026](#)).

COSIMA Cookbook

Search Go

Navigation

Lessons

Cooking Lessons 101

- Basics
- Advanced

Recipes

Appetisers

Mains

Local Dishes

Resources

Contributing

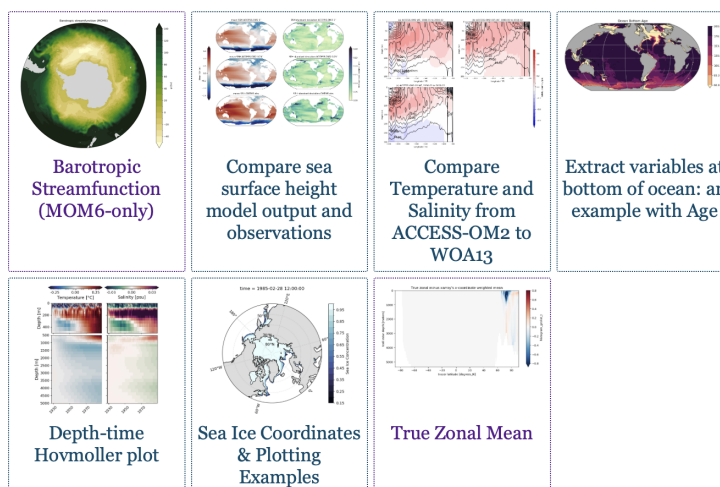
- Getting started with git
- Working on your recipe
- Tips for making a good recipe
- Submitting a Pull Request
- Reviewing existing Pull Requests

Notebook Guidelines

- Thumbnails
- Images in notebooks

GitHub Repository

Appetisers (easy)



latest

Figure 1: The COSIMA Cookbook documentation website, showing the browsable Sphinx-based landing page used to navigate tutorials and recipes. The live site is available at <https://cosima-recipes.readthedocs.io>.

The instructional structure is central to the project. “Cooking Lessons 101” introduces generic skills such as loading model output, working with labelled arrays, plotting, and interacting with shared data catalogues. These tutorials lead into domain-focused “recipes”: self-contained notebooks showing how to perform concrete diagnostics and analyses on ocean and sea-ice datasets. The current repository contains dozens of notebooks grouped into introductory, advanced, and region-specific collections, giving learners an incremental path from first contact with the data ecosystem to adaptation of full workflows for their own science questions.

The Cookbook grew from a practical need inside the COSIMA community: the tools used to analyse modern ocean model output are powerful, but the gap between package-level documentation and a reproducible end-to-end workflow remains large. Users need to understand not only Python and Jupyter (Kluyver et al., 2016), but also how to navigate high-dimensional model output, shared high-performance computing environments, and domain-specific analysis conventions. The Cookbook addresses that gap by packaging reusable workflows in the same medium in which users actually work.

Statement of Need

Ocean and climate model analysis has a steep entry cost. New users must learn how to find datasets, load them efficiently, interpret metadata, operate on multi-dimensional arrays, and produce scientifically meaningful diagnostics. General-purpose libraries such

51 as xarray (Hoyer & Hamman, 2017) provide the foundations for these tasks, but they
52 do not by themselves show a learner how to translate a research question into a robust
53 analysis workflow for a specific model and computing environment.

54 The COSIMA Cookbook fills that need with a domain-specific, openly maintained collection
55 of computational lessons and examples. Its contribution is not a new analysis package;
56 rather, it is a reusable learning resource that makes existing scientific software, data
57 conventions, and model output more approachable. This aligns well with JOSE's emphasis
58 on open educational resources that enable computational learning through authentic
59 practice.

60 Several design decisions make the Cookbook particularly useful for adoption by other groups.
61 First, each notebook is intended to be self-contained and well-documented, so learners
62 can read, run, and modify a complete workflow without reconstructing missing context
63 from scattered notes. Second, the repository distinguishes tutorials from recipes. Tutorials
64 teach transferable skills, while recipes demonstrate how those skills combine in realistic
65 analysis tasks. Third, the project includes contributor guidance and notebook review
66 conventions that encourage new analyses to be written in a reusable and pedagogically
67 clear style.

68 This structure is valuable beyond the immediate COSIMA context. Many research
69 communities maintain model output on shared infrastructure and face the same challenge:
70 turning expert tacit knowledge into examples that newcomers can adapt. The Cookbook
71 offers a model for how a scientific collaboration can capture that knowledge in version-
72 controlled notebooks, publish it as a living resource, and continuously improve it through
73 community contribution.

74 Educational Design and Experience of Use

75 The learning experience is organised as a progression. The documentation points new users
76 first to introductory lessons, including material on loading, slicing, and visualising model
77 output. Learners then move to more advanced tutorials and finally to recipe notebooks
78 that address concrete scientific questions. The categories of “appetisers”, “mains”, and
79 “local dishes” provide a lightweight pedagogical cue about expected complexity and scope.

80 The intended mode of use is also explicit. The repository is designed around Jupyter-
81 based analysis on the Australian National Computational Infrastructure, with guidance
82 on running notebooks through ARE/JupyterLab sessions and on accessing shared data
83 holdings. This makes the Cookbook more than a static collection of examples: it is
84 operational documentation for a real analysis environment. At the same time, the
85 notebooks demonstrate broadly transferable patterns for working with labelled geophysical
86 data, so individual lessons can be reused or adapted outside that environment.

87 The project also supports community authorship. Contributors are encouraged to submit
88 new notebooks through pull requests, document their methods clearly, and generalise
89 workflows so that they remain useful to others. In that sense, the Cookbook functions
90 both as a learning module for end users and as a framework for teaching reproducible
91 scientific communication through notebook design and peer review.

92 Author order

93 The first two authors contributed equally; authors from the third position onward are
94 listed alphabetically by surname.

95 Acknowledgements

96 We acknowledge the COSIMA community for developing and maintaining the model output,
97 workflows, and pedagogical practices reflected in this repository. We also acknowledge
98 the National Computational Infrastructure, which is supported by the Commonwealth
99 government of Australia, and the ACCESS-NRI ecosystem that make the operational
100 use of these notebooks possible. We acknowledge funding from the Australian Research
101 Council under the Center of Excellence for the Weather of the 21st Century CE230100012,
102 the Linkage Infrastructure, Equipment and Facilities LP200100406, and the Discovery
103 Project DP240101274.

104 References

- 105 COSIMA contributors. (2026). *COSIMA cookbook: A cookbook of recipes for analysing*
106 *ocean and sea ice model output*. <https://doi.org/10.5281/zenodo.14353852>
- 107 Hoyer, S., & Hamman, J. J. (2017). Xarray: N-D labeled arrays and datasets in Python.
108 *Journal of Open Research Software*, 5(1), 10. <https://doi.org/10.5334/jors.148>
- 109 Kiss, A. E., Woodward, M. J., Hollings, T., Mu, M., Fiedler, R., Steven, A. D. L., Lenton,
110 A., Matear, R., Chamberlain, M. A., Oke, P. R., Alves, O., Griffies, S. M., Hill, C.,
111 Kovach, A., Marsland, S. J., Fiedler, R., Waters, J., Yang, C.-H., Zhou, X., ... Hogg, A.
112 McC. (2020). ACCESS-OM2 v1.0: A global ocean-sea ice model at three resolutions.
113 *Geoscientific Model Development*, 13(2), 401–442. [https://doi.org/10.5194/gmd-13-](https://doi.org/10.5194/gmd-13-401-2020)
114 [401-2020](https://doi.org/10.5194/gmd-13-401-2020)
- 115 Kluyver, T., Ragan-Kelley, B., Pérez, F., Granger, B., Bussonnier, M., Frederic, J.,
116 Kelley, K., Hamrick, J., Grout, J., Corlay, S., Ivanov, P., Avila, D., Abdalla, S.,
117 & Willing, C. (2016). Jupyter notebooks—a publishing format for reproducible
118 computational workflows. In F. Loizides & B. Schmidt (Eds.), *Positioning and power*
119 *in academic publishing: Players, agents and agendas* (pp. 87–90). IOS Press. [https:](https://doi.org/10.3233/978-1-61499-649-1-87)
120 [//doi.org/10.3233/978-1-61499-649-1-87](https://doi.org/10.3233/978-1-61499-649-1-87)